

Digital Signal Generator and Line Coding Visualizer

ITT-036 Programming Assignment 2025

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1. Objective

The aim of this project is to design and implement a **Digital Signal Generator** that supports both analog-to-digital conversion (using **Pulse Code Modulation (PCM)** or **Delta Modulation (DM)**) and digital line coding techniques (**NRZ-L**, **NRZ-I**, **Manchester**, **Differential Manchester**, **AMI**) with optional scrambling (**B8ZS**, **HDB3**). The system also visualizes the generated digital waveform and analyzes important signal properties such as the **longest palindrome** and **longest zero sequence** within the data stream.

2. Programming Language and Libraries

- **Language:** C++17
- **Platform:** macOS (tested on macOS Sequoia / Sonoma)
- **Graphics Library:** OpenGL (macOS framework) and GLUT (<GLUT / glut.h>)
- **Compiler:** clang++ / g++ (Xcode Command Line Tools)

3. Features Implemented

- User-selectable **digital** or **analog** input
- **PCM** and **Delta Modulation** for analog-to-digital conversion
- Line encoding techniques: **NRZ-L**, **NRZ-I**, **Manchester**, **Differential Manchester**, **AMI**
- **Scrambling** schemes: **B8ZS** and **HDB3**
- **Longest palindrome detection** using **Manacher's Algorithm** ($O(n)$ time complexity)
- **Longest zero-sequence detection** for competitive optimization
- **Real-time waveform visualization** using OpenGL on macOS

4. Assumptions and Design Considerations

- Input digital stream contains only **0s and 1s**
- Analog sample values are floating-point between **-1.0 and +1.0**
- OpenGL window automatically scales based on signal length
- Scrambling is applied **only** for **AMI encoding**
- Manchester and Differential Manchester double the visual signal length

5. How to Run the Code (macOS)

1. Save your file as: **SignalSimulator.cpp**
2. Open Terminal and compile using:

```
g++ -std=c++17 SignalSimulator.cpp -o SignalSim \
-framework OpenGL -framework GLUT -DGL_SILENCE_DEPRECATION
```
3. **./SignalSim**

7. Acknowledgement

This project was developed independently by the author. Discussions with classmates helped in understanding waveform visualization and PCM quantization.

Concepts were inspired by Dr. Iqra ma'am course of data communication and Apple's OpenGL documentation. No direct code was copied; all program structures were designed and written from scratch to ensure originality.